Reconstruction of the Dimuon Invariant Mass Spectrum from CMS Open Data at $\sqrt{s} = 8\,\text{TeV}$

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Abstract

I reconstruct the dimuon invariant mass spectrum from about 61,5 million proton collision events recorded by the CMS experiment at the LHC (Large Hadron Collider) in 2012 at a centre of mass energy of 8 TeV, using openly available data released through the CERN Open Data Portal. Selecting events with exactly two oppositely charged muons yielded approx 24,1 million usable events, in which key known resonances from the η meson to the Z boson are clearly observed at their known masses. My fit to the Z peak ends up giving 90.729 GeV with an uncertainty of +/- 0.001 GeV, reaching within 0.50% uncertainty. This deviation does not decrease from 1,000,000 events all the way up to approx 61,500,000 events (pre-filtering), showing that the error is systematic. This could be from a number of factors including certain background radiation perhaps or calibration issues.

1 Introduction

Most particles produced in high-energy collisions are unstable and decay long before they can be detected directly. Their properties therefore must be inferred from their **decay products**. Pairs of oppositely charged muons are well suited to be observed since they penetrate the **outer layer** of the detector where almost all other particles are - by this point - absorbed, making them easy to identify and measure precisely. The invariant mass of a muon pair has in the pass enabled discovery of numerous particles, and is what allows me to rediscover particles using real world data.

2 Theory

The invariant mass of a muon pair is

$$m_{\mu\mu}^2 = (E_1 + E_2)^2 - |\vec{p}_1 + \vec{p}_2|^2. \tag{1}$$

This quantity is the same in every reference frame, so it equals the parent particle's mass regardless of how fast the parent was moving. The CMS describes each muon by its transverse momentum p_T (the momentum perpendicular to the beam), pseudorapidity η (related to the polar angle from the beam axis) and azimuthal angle ϕ . The Cartesian momentum components and energy are as follows:

$$p_x = p_T \cos\phi, \qquad p_y = p_T \sin\phi, \qquad p_z = p_T \sinh\eta \tag{2}$$

$$E = \sqrt{p_x^2 + p_y^2 + p_z^2 + m_\mu^2} \tag{3}$$

Where $m_\mu = 0.106$ GeV is the muon mass. An unstable particle does not have a single ... CONTENT TO BE CONTINUED SOON (jupyter notebook fully works)

3 Data and Detector

4 Method

5 Results

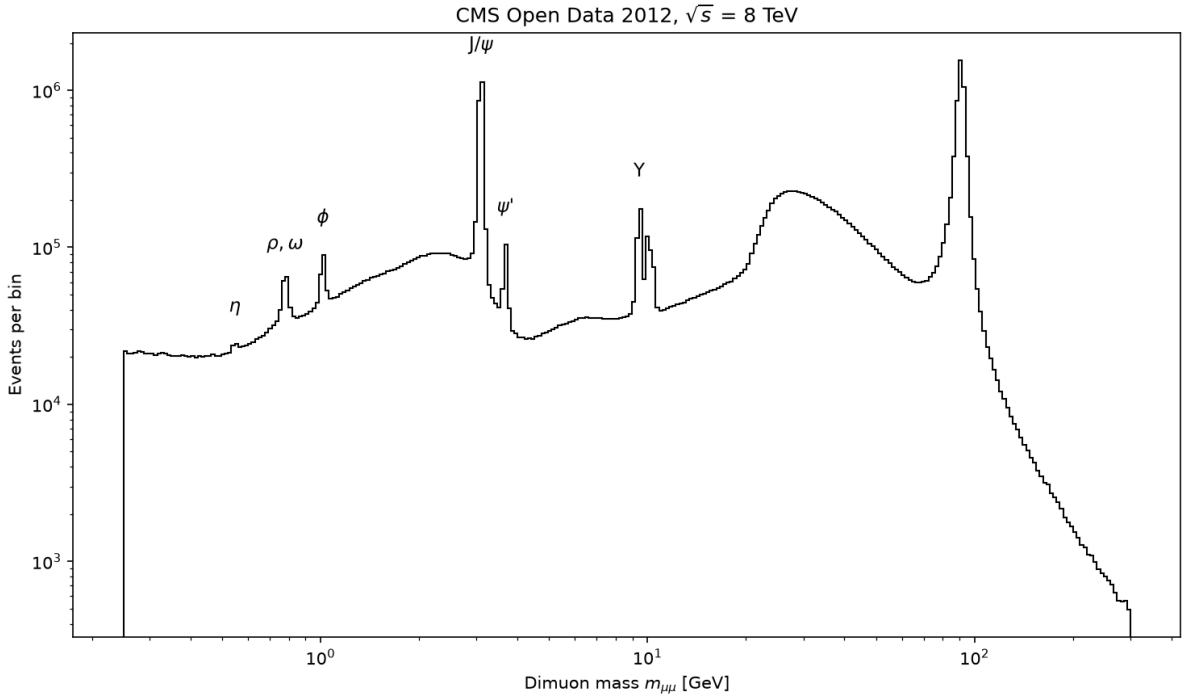


Figure 1: Dimuon invariant mass spectrum for 24,067,843 opposite-charge muon pairs. Known resonances are labelled.

6 Discussion

7 Conclusion

References

[1] CMS Collaboration, DoubleMuParked dataset (Run2012B/C), CERN Open Data Portal, DOI: [from record page].

[2] Particle Data Group, *Review of Particle Physics*.